# Anomaly Localization using Diffusion Models

Group 4
Vozinaki Anthi 03119168
Giadikiaroglou Panagiotis 03119185
Karafyllis Nikolas 03119890
Litsos Ioannis 03119135
Michelakis Panagiotis 03119119
School of Electrical and Computer Engineering
National Technical University of Athens
Pattern Recognition
Athens, Greece

***Abstract*—This study builds upon the work of Sanchez et al. [13], which introduced the use of diffusion models for the detection and localization of brain anomalies, by not only reconstructing their research to validate findings but also extending it through innovative experimentation. Our approach involved adjusting various hyperparameters within the diffusion models and applying these modified models to a new dataset to assess their robustness and generalizability. Our findings confirm the efficacy of diffusion models in accurately detecting brain anomalies, as initially reported, and demonstrate that through careful optimization of hyperparameters, the performance of these models can be significantly enhanced. The successful application to a different dataset further validates the model's adaptability and potential for broader clinical application, suggesting a promising avenue for future research in medical imaging and the early diagnosis of neurological disorders.**

***Index Terms*—Generative Models, Diffusion Models, DDPMs, SMLDs, SDEs, DDIM, LDMs, Unconditional Generation, Guidance, Anomaly detection, Brain.**

## I. Introduction

Generative models have gained much popularity in recent years due to their ability to generate high-quality images, synthesize music and human-like speech. Among the various generative models that have been proposed over the years - GANs [5], VAEs[8], autoregression transformers, flow models - diffusion models [2, 6, 15, 14, 16, 17, 18] have been consistently producing the best results among various fields [3, 10, 12]. By leveraging their generative capabilities, we can synthesize counterfactuals [1] of *"How would a patient's brain appear if their pathology was not present?"*. The discrepancy image created by comparing the observed patient condition with a healthy counterpart serves as a basis for identifying the pathology's location. Counterfactuals are produced to represent the smallest necessary alteration of the input to transition it into a healthy state.

## II. Theoretical Background behind Diffusion Models

Sohl-Dickstein et al. [14] set the general framework for the diffusion process in 2015. However, they left out many important details regarding the implementation of said process. This theoretical and practical gap was filled by another seminal paper in the field; ***"Denoising Diffusion Probabilistic Models"*** [6]. This paper introduced numerous new concepts expanding on the work of [1]. Issues such as variance schedule, a formal definition of the DDPM sampling procedure, and a more tractable definition of the cost were discussed.

The *forward process* is described by the following Markovian process:

$$q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{1-b_t} \cdot x_{t-1}; b_t \cdot I), \forall t \in 1, ..., T \quad (1)$$

where **T** is the number of diffusion steps, $b1, ..., b_T \in [0, 1)$ is the variance schedule , **I** is the identity matrix that has the same dimensions as the input image $x_0$, and $\mathcal{N}(x; \mu, \sigma)$ represents the normal distribution of mean $\mu$ and covariance $\sigma$ that produces x.

In the reverse process, we start from a sample $x_T \sim \mathcal{N}(0, I)$, and generate new samples from $p(x_0)$ by following the reverse steps:

$$p(x_{t-1}|x_t) = \mathcal{N}(x_{t-1}; \mu(x_t, t), \Sigma(x_t, t)) \quad (2)$$

In order to approximate the reverse steps, a neural network is trained that receives a noise input and learns to predict the mean and covariance.

$$p_\theta(x_{t-1}|x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t)) \quad (3)$$

Given that both of the forward and the reverse processes are modeled as Markov chains, the joint probability distribution in each case can be written as follows:

$$q(x_1, x_2, ..., x_T|x_0) = \prod_{t=1}^{T} q(x_t|x_{t-1}) \quad (4)$$

$$p_\theta(x_1, x_2, ..., x_T) = p(x_T) \prod_{t=1}^{T} p_\theta(x_{t-1}|x_t) \quad (5)$$

Equivalently, regarding the actual perturbed data, the discrete Markov chain that describes the forward diffusion process can be expressed as follows:

$$x_t = \sqrt{1 - b_t} x_{t-1} + \sqrt{b_t}\epsilon \tag{6}$$

where

$$x_0 \sim p(x_0)$$
$$\epsilon \sim \mathcal{N}(0, I)$$

There is also a transformation that can be used in order to obtain a sample $x_t$ directly from the initial sample $x_0$.

$$x_t = \sqrt{\bar{a_t}} x_0 + \sqrt{1 - \bar{a_t}}\epsilon \tag{7}$$

where

$$a_t = 1 - b_t$$
$$\bar{a_t} = \prod_{s=0}^{t} a_s$$

The reverse process is described by the following equation:

$$x_{t-1} = \mu_\theta(x_t, t) + \sqrt{\Sigma_\theta(x_t, t)} z \tag{8}$$

where

$$x_T \sim \mathcal{N}(0, I)$$
$$z \sim \mathcal{N}(0, I)$$

It is noted that $\epsilon$ are standard spherical Gaussian noise terms independent of the *past* values of x and $z$ are standard spherical Gaussian noise terms independent of the *future* values of x. Both of these terms depend on time t, i.e. $\epsilon = \epsilon_t$, $z = z_t$, but we avoid the subscripts in favor of simplicity.

Notice how during the reverse process, additional noise is added to the predicted sample at every time step. This is to ensure that the generation process does not get stuck in modes of the distribution.

For our model to produce accurate and high-quality results, we want the reverse joint probability distribution, $p_\theta(x_0, x_1, ..., x_T) = p(x_T) \prod_{t=1}^{T} p_\theta(x_{t-1}|x_t)$ to closely approximate the joint forward probability distribution, $q(x_0, x_1, ..., x_T) = q(x_0) \prod_{t=1}^{T} q(x_t|x_{t-1})$. For this reason, the minimization of the Kullback-Leibler (KL) divergence between these two distributions is used as the objective for training the neural network:

$$KL(q(x_0, x_1, ..., x_T)||p_\theta(x_0, x_1, ..., x_T)) \tag{9}$$

$$= L_{VLB} + const \tag{10}$$

$$\geq \mathbb{E}[-log p_\theta(x_0)] + const \tag{11}$$

The term $L_{VLB}$ represents the Variational Lower Bound of the log-likelihood of the data $x_0$. This is a commonly used objective in neural network training since it closely approximates the log-likelihood of the data and is computationally tractable. The term $const$ represents a constant that does not depend on the network's parameters, $\theta$.

After some simplification, the authors arrive at the following form for the $L_{VLB}$:

$$\begin{aligned}\mathbb{E}_{q(x_0, x_1, ..., x_T)} \big[ & KL(q(x_T|x_0)||p(x_T)) \\ & + \Sigma_{t=1}^{T} KL(q(x_{t-1}|x_t, x_0)||p_\theta(x_{t-1}|x_t)) \\ & - \log p_\theta(x_0|x_1)\big]\end{aligned} \tag{12}$$

The distribution $q(x_{t-1}|x_t, x_0)$ is called the ***forward process posterior distribution***:

$$q(x_{t-1}|x_t, x_0) = \mathcal{N}(x_{t-1}; \tilde{\mu}_t(x_t, x_0), \tilde{b_t} I) \tag{13}$$

where:

$$\tilde{\mu}_t(x_t, x_0) = \frac{\sqrt{\tilde{a}_{t-1}} b_t}{1 - \tilde{a}_t} x_0 + \frac{\sqrt{a_t}(1 - \tilde{a}_{t-1})}{1 - \tilde{a_t}} x_t \tag{14}$$

$$\tilde{b}_t = \frac{1 - \tilde{a}_{t-1}}{1 - \tilde{a}_t} b_t \tag{15}$$

It can also be proven that minimizing the KL divergence between two Gaussian distributions when the variance is fixed, is equivalent to minimizing the distance between their means. In this case, the aforementioned training objective can be further simplified to the following:

$$L_{t-1} = \mathbb{E}_{q(x_0, x_1, ..., x_T)} \left[ \frac{1}{2\sigma_t^2} ||\tilde{\mu}_t(x_t, x_0) - \mu_\theta(x_t, t)||_2^2 \right] + const, \tag{16}$$

where $\sigma_t^2 = b_t$.

Finally, we can express $x_0$ as a function of $x_t$ from equation (5) and acquire a simplified version of $\tilde{\mu}_t$:

$$\tilde{\mu}_t(x_t, x_0) = \tilde{\mu}_t(x_t) = \frac{1}{\sqrt{a_t}} \left( x_t - \frac{b_t}{\sqrt{1 - \tilde{a}_t}} \epsilon \right) \tag{17}$$

Now, since we want $\mu_\theta \approx \tilde{\mu}_t$ and $x_t$ is already available to the input model, the simplified objective becomes:

$$L_{simple} = \mathbb{E}_{t, x_0, \epsilon} \left[ ||\epsilon - \epsilon_\theta(x_t, t)||_2^2 \right] \tag{18}$$

Equivalently, using equation (7), the above objective can be written:

$$L_{simple} = \mathbb{E}_{t, x_0, \epsilon} \left[ ||\epsilon - \epsilon_\theta(\sqrt{\tilde{a}_t} x_0 + \sqrt{1 - \tilde{a}_t}\epsilon, t)||_2^2 \right] \tag{19}$$

The model now tries to predict the added noise at each step in the forward diffusion process. With the variance matrix fixed and the noise $\epsilon_\theta$ of each time step available, we can sample from the initial distribution using equation (6):

$$x_{t-1} = \frac{1}{\sqrt{a_t}} \left[ x_t - \frac{1 - a_t}{\sqrt{1 - \tilde{a}_t}} \epsilon_\theta(x_t, t) \right] + \sigma_t z \qquad (20)$$

## III. Original Paper Reconstruction and Results

Our goal is to alter the input image from an unhealthy to a healthy state during the inference phase, while keeping all other elements of the image constant. The focus is on determining which key feature needs to be changed. For example, in a brain tumor dataset [9], the primary features to be adjusted might include lesions. This process of making the smallest necessary change is referred to as counterfactual generation [1] in causal research. After training our model on imaging data where $c$ falls within the range of *[healthy, unhealthy]*, we are able to shift the input image from one domain to another at inference time through counterfactual generation.

In more detail, encoding is done by iteratively applying diffusion models to obtain a latent space [12] with an unconditional model $(c = 0)$ [3, 7]. Decoding is performed by reversing the diffusion process from the latent space to reconstruct an image with healthy condition. Similarly, decoding uses *healthy* and *0* for guidance. A heatmap of the lesion can obtained by taking the difference between the original and the reconstructed healthy. The aforementioned procedure is represented in the following picture:

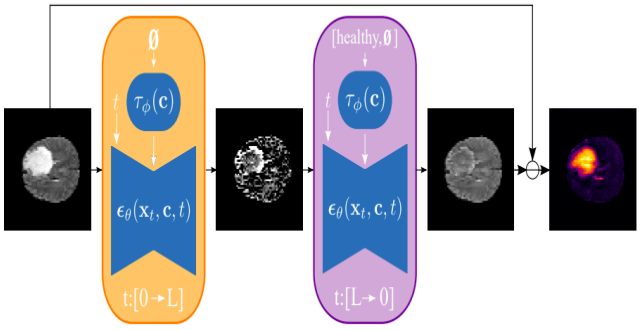



Fig. 1. Counterfactual generation with diffusion models overview

Based on the proposed method, we were able to reproduce the paper's results regarding the localization of anomalies in brain MRIs. Some of these results are presented below.

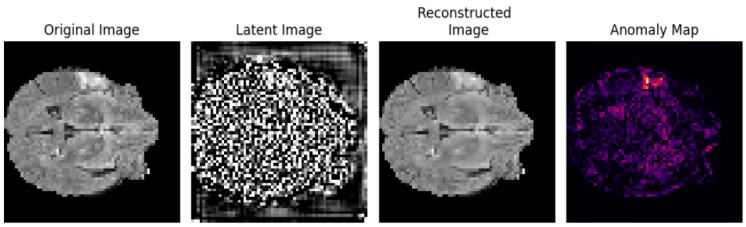



Fig. 2. Brain Anomaly Localization - Result 1

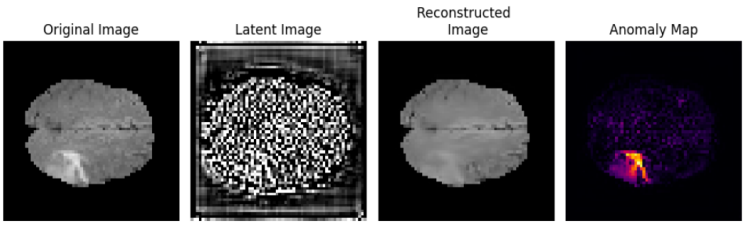



Fig. 3. Brain Anomaly Localization - Result 2

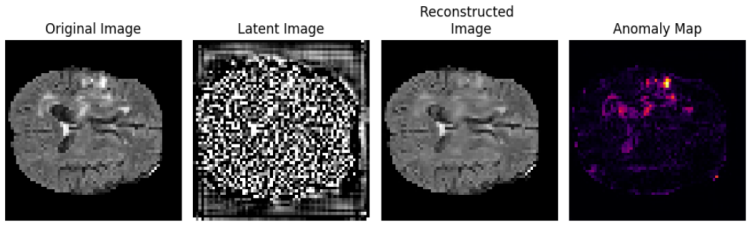



Fig. 4. Brain Anomaly Localization - Result 3

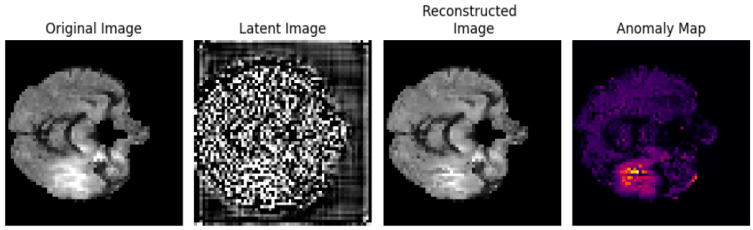



Fig. 5. Brain Anomaly Localization - Result 4

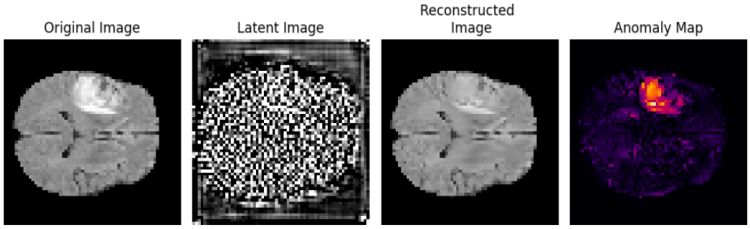



Fig. 6. Brain Anomaly Localization - Result 5

For comparison, we calculated the **Dice score** [4] for our generated results. In the original paper, Sanchez et al. [13] managed to acquire an average Dice score of **76.2**. For our calculation of the Dice score, we utilized both the **standard binary** mask (manual threshold) as well as the **Otsu threshold mask** [11]. We acquired average Dice scores of **61.4** and **63.4** respectively. Evidently, we produced results of lower quality and accuracy than the original paper but still respectable according to the general consensus. The difference in quality can be attributed to several factors including the smaller size of our training dataset, the shorter training for our model as well as the limited computational resources that were available. We present below an example of the binary and Otsu masks we produced for an image of the validation dataset in our experiment.

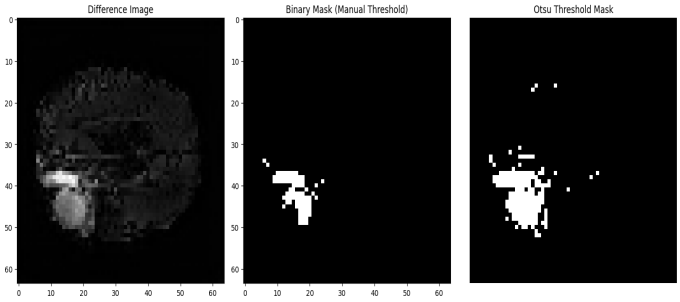



Fig. 7. Binary and Otsu masks of the difference image

## IV. Further Experimentations

As an initial modification to the original study, we explored the adjustment of various hyper-parameters. In particular, we experimented with the values of *training iterations*, *learning rate*, and *number of timesteps* in the sampling process. Some of our results are presented below:

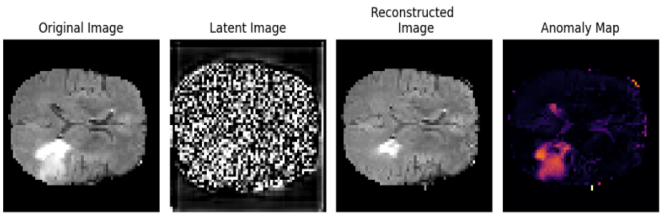



Fig. 8. training iterations = 5e3, lr=1e-3, total timesteps = 1000

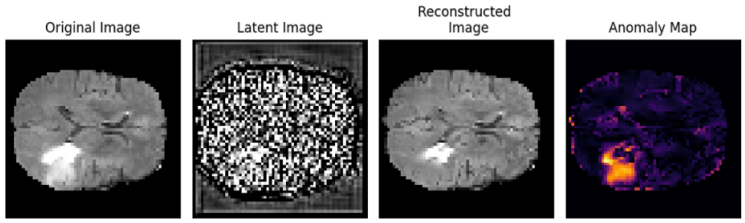



Fig. 9. training iterations = 5e3, lr=1e-4, total timesteps = 1000

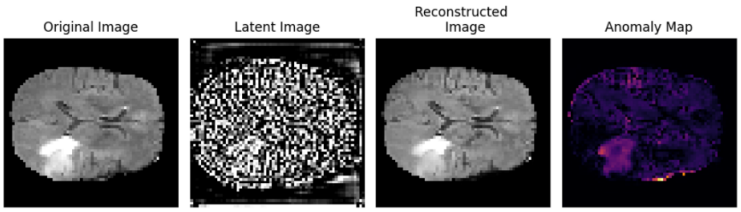



Fig. 10. training iterations = 5e3, lr=1e-5, total timesteps = 1000

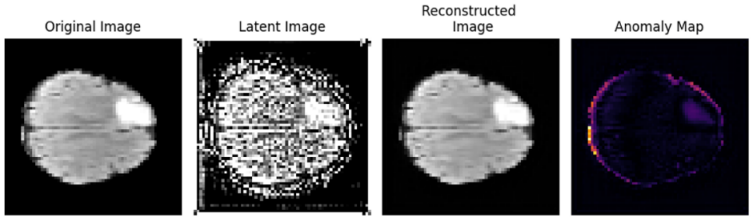



Fig. 11. training iterations = 2.5e3, lr=1e-5, total timesteps = 250

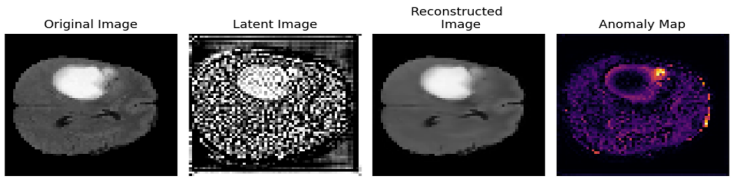



Fig. 12. training iterations = 1e4, lr=1e-5, total timesteps = 250

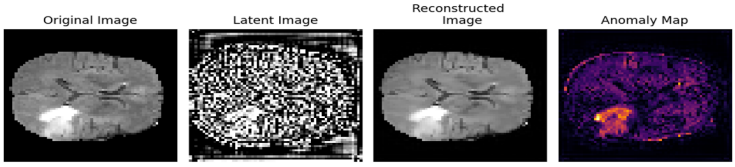



Fig. 13. training iterations = 5e3, lr=1e-5, total timesteps = 250

In the original paper, [13] as well as in our own experiment, the hyper-parameters were set to the following values:

- **training iterations = 20000**
- **learning rate = 1e-5**
- **total timesteps = 1000**

Based on our experimentation with the hyper-parameters, some patterns emerged regarding each one's relevance in the process:

- Reducing the number of total timesteps in the sampling process results in lower-quality anomaly map images.
- Training iterations don't change the model's performance if they exceed 5000.
- Learning rates of the order e-4 and e-5 produced the best overall results. Learning rates of the order e-3 and higher produced sub-optimal results.

## V. Anomaly Detection in Liver

As part of our study, we aimed to evaluate the performance of our model on another dataset. For this purpose, we once again focused on leveraging the Decathlon Dataset, now targeting the liver segmentation task. This dataset consists of 201 contrast-enhanced CT images sourced from patients diagnosed with primary cancers and metastatic liver diseases, associated with colorectal, breast, and lung primary cancers. Among these images, 105 were allocated for training the model, while 26 were reserved for validation purposes.

For the hyperparameters, we used

- **training iterations = 10000**
- **learning rate = 1e-5**
- **total timesteps = 1000**

.

Although our model successfully detects certain anomalies, overall, its performance is inferior to that achieved on the BRATS dataset. This disparity can be attributed to several factors: 1. *Limited Data Availability*: The dataset used for liver segmentation contains fewer samples compared to the BRATS dataset, potentially limiting the model's ability to generalize effectively. 2. *Structural Consistency*: The brain exhibits a relatively consistent structure and shape across individuals, resulting in less variability compared to the liver. 3. *Increased Anomaly Diversity*: Liver anomalies demonstrate greater diversity in terms of size, shape, and appearance compared to those found in brain imaging. 4. *Higher image resolution*: Liver task contains images with higher resolution than the BRATS dataset. This means it demands more computational resources, including memory and processing power.

Below, we present select outputs generated by our model for the liver segmentation task:

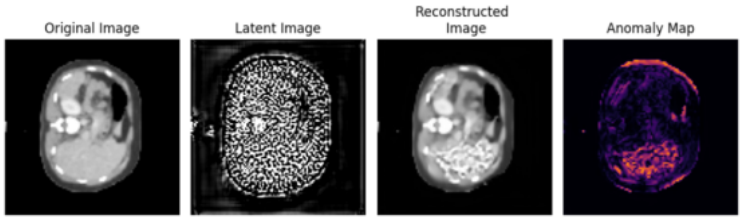



Fig. 14. Liver Anomaly Localization - Result 1

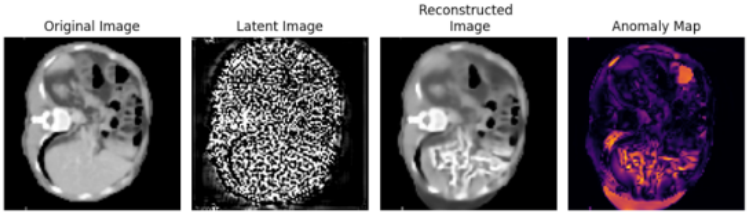



Fig. 15. Liver Anomaly Localization - Result 2

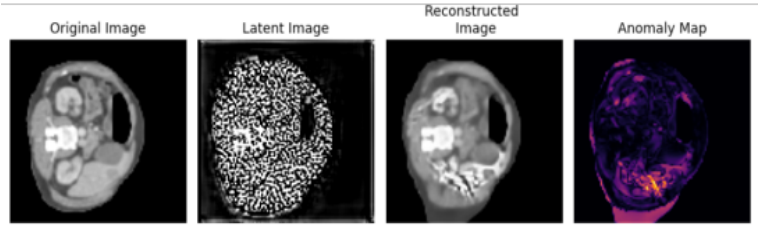



Fig. 16. Liver Anomaly Localization - Result 3

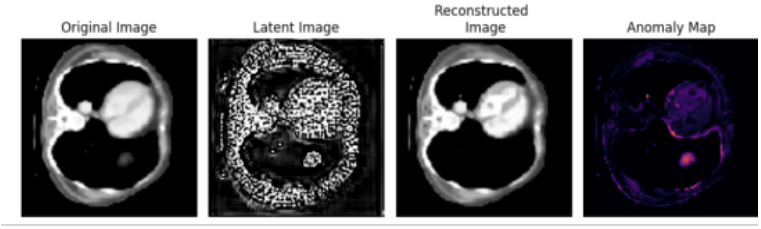



Fig. 17. Liver Anomaly Localization - Result 4

The average **Dice** score for the Liver Anomaly Localization task was found to be **0.51**. This is a mediocre performance on this dataset caused by the reasons we previously mentioned.

## VI. Final Remarks

Our work has successfully demonstrated the efficacy of using diffusion models for generating counterfactuals to detect and localize anomalies in brain imaging, thereby reconstructing and validating the pioneering research conducted by Sanchez et al [13]. By meticulously analyzing the difference images between the original and counterfactual representations, we have highlighted the potential of this approach in identifying the precise locations of brain anomalies. Moreover, our contributions extend beyond mere replication; we experimented with the model's hyper-parameters, which provided us with deeper insights into optimizing and analyzing the model's performance. Furthermore, the application of this model to a different dataset—focusing on liver anomaly localization underscores the versatility and robustness of our approach. The success in both brain and liver anomaly detection exemplifies the potential of diffusion models as a powerful tool in medical imaging, opening new avenues for research and clinical applications. Our findings not only reinforce the importance of counterfactual generation in medical diagnostics but also pave the way for future studies to explore the broad applicability of diffusion models across various domains of medical imaging.

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